Abubakar Sadiq Shittu

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Education

Ph.D. Computer Science, *University of Tennessee, Knoxville* — Jan 2025 – Anticipated: Dec 2027

GPA: 3.67/4.00 | Coursework: Advanced Software Engineering, Algorithms, Data-Driven Security & Privacy, Computer Systems Organization, Digital Archaeology.

M.Sc. Advanced Computer Science, *Newcastle University, United Kingdom* — Sept 2023 – Sept 2024

GPA: 4.00/4.00 (Distinction) | Coursework: Machine Learning, Secure Software Development, Systems Security, Risk & Trust Management.

B.Sc. Computer Science, *Fountain University, Osogbo* — March 2018 – June 2021

GPA: 4.63/5.00 (First Class Honours)

Experience

Graduate Research Assistant — Jan 2025 – Present

Usable Security and Engineering Research Laboratory, Led by Dr. Scott Ruoti — *United States*

- Architected a high-throughput, containerized data ingestion pipeline using Docker; integrated GitHub REST and GraphQL APIs with a PostgreSQL backend to continuously harvest 4M+ user profiles and 20M+ commit records [1].
- Spearheaded longitudinal mixed-methods studies on (1) TLS client authentication, triangulating participant feedback, server logs, and 400+ certificate signing requests to identify usability failures [2]; and (2) git commit signing, validating qualitative findings against `.git` log data and usability benchmarks (SUS/ASQ) to pinpoint technical and usability barriers to its adoption as a supply chain defense [manuscript in progress].

Technical and Administrative Assistant — Oct. 2021 – Oct. 2022

Kaduna State University — *Nigeria*

- Served as the primary technical point of contact for the *examination and records unit*, diagnosing and resolving hardware, software, and network connectivity issues to ensure operational continuity.

Information Technology Helpdesk Intern — Mar. 2019 – Aug. 2019

Nigerian National Petroleum Corporation — *Nigeria*

- Supported 100+ computer users as the primary technical point of contact, resolving 5+ daily issues.

Preprints

[1] **A. S. Shittu**, J. Sadik, F. Gholamrezae, and S. Ruoti, "Analysis of commit signing on GitHub," *arXiv preprint arXiv:2604.14014*, 2026. [Online]. Available: https://arxiv.org/pdf/2604.14014.

[2] **A. S. Shittu**, C. Shubert, J. Sadik, and S. Ruoti, "Understanding student experiences with TLS client authentication," *arXiv preprint arXiv:2604.14330*, 2026. [Online]. Available: https://arxiv.org/pdf/2604.14330.

Selected Class Projects

Machine Learning Pipelines, Newcastle University, UK — Dec 2023

- Built end-to-end pipelines for classification, regression, and NLP clustering; tuned Random Forest on FARS (100K+ records) improving ROC-AUC 88.05% → 93.09%; benchmarked SVM, Gradient Boosting (90.50%), and Logistic Regression (93.25%) with SMOTE; predicted bacterial growth parameters (μ, τ, a_0) across 19,992 experiments; compared TF-IDF vs. embeddings for K-Means clustering on 2,225 news articles; and tuned an LSTM on Raspberry Pi sensor data achieving $R^2 = 0.95$.

Advanced RAG Pipeline — Spring 2025

- Built a RAG QA pipeline (LangChain + Gemini) with RecursiveCharacterTextSplitter chunking and VertexAIEmbeddings for retrieval; scraped and normalized a university course-schedule (BeautifulSoup) into a retrieval-optimized JSON dataset; indexed records as LangChain Documents in a vector store to serve natural-language queries.

Scholarships and Awards

- **Travel Grant**, *Summer School on Real-World Crypto and Privacy, Vodice, Croatia* — 2024
- **Chevening Scholarship**, *United Kingdom. ∼$60,000: full tuition, monthly stipend, and travel* — 2023
- **Commonwealth Shared Scholarship** (Declined), *United Kingdom. ∼$60,000* — 2023
- **100% Scholarship** (Declined) – *M.Sc. Computer Engineering, Kadir Has University, Turkey* — 2022
- **Most Outstanding Student**, *College of Natural and Applied Sciences, Fountain University* — 2019

Skills, Service, and Affiliations

- **Development:** Python, C/C++, Docker, SQL, PostgreSQL, REST/GraphQL APIs | **Systems & Analysis:** macOS, Windows, Linux/Unix, Bash, SonarQube, Wireshark, Git.
- **Reviewing:** Journal of Cybersecurity, Q1 (SJR), 2025 | **Affiliations:** Graduate Student Member, IEEE & IEEE Computer Society | **Volunteering:** Tennessee Extension Volunteer | **User Research:** NVivo, MAXQDA, Survey, Interview.